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Buckling analysis of porous functionally graded GPL-reinforced conical shells subjected to combined forces

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Abstract This paper presents the analytical process of porous functionally graded graphene-reinforced truncated conical shells subjected to hydrostatic pressure and axial tension. Three types of graphene platelet (GPL) dispersion and three patterns of porous distribution were considered. A modified model, in which the volume fraction of the pores is regarded as the essential parameter, was built to evaluate the material properties. The static stability equations of the shells, coupled with the effect of the combined forces, were derived. The Galerkin integrate technique was employed to obtain the critical buckling hydrostatic pressure and axial tension. After the present method was validated, the influences of pores, GPLs, and shell geometrical characteristics were investigated in the parametric studies. The results show that the critical hydrostatic pressure and axial tension can be elevated by increasing the porous coefficient, semi-vertex, and length–thickness ratio. On the contrary, the critical pressure and tension are decreased with the rise of GPL mass fraction.

Keywords Buckling · Conical shell · Graphene platelet · Porous · Combined forces

1 Introduction

Truncated conical shells are used in many engineering fields, such as aerospace, marine, civil, and mechanical engineering. If these shells are subjected to high compressive forces, a buckling phenomenon may take place [1]. Therefore, it is necessary to study the buckling characteristics of the shells for a safe design.

In the past decade, many studies have been conducted on the buckling and vibration of functionally graded structures subjected to mechanical [2–7], thermal [8–12], and electromagnetic [13, 14] loads. However, research work on the functionally graded shell structures reinforced with carbon nanotubes (CNTs) and graphene platelets (GPLs) still needs to be improved. Ansari and Torabi [15] and Hosseini and Talebitooti [16] studied the buckling behavior of functionally graded carbon nanotube-reinforced composite (FG-CNTRC) conical shells under axial pressure by using the generalized differential quadrature method and differential quadrature method, respectively. In their parametric studies, it was proved that the buckling pressure can be raised by increasing the volume fraction of CNTs. Employing the Galerkin method, Duc et al. [17] analyzed the thermal

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and mechanical stability of FG-CNTRC truncated conical shells surrounded by elastic foundations. They found that the volume fraction and distribution patterns of fibers, semi-vertex angle, shell length, temperature, and the elastic foundation can significantly affect the thermal and mechanical buckling loads. Applying the Galerkin method, Sofiyev and his coauthors [18–20] obtained the analytical solution of the stability for the FG-CNTRC conical shells subjected to external and combined loads and investigated the influences of the geometrical parameters of the shells and the volume fraction and distribution types of CNTs. They found that the buckling load increases with the rising volume fraction of CNTs. Moreover, the increasing trend declines when the nanotubes in the shell are saturated. Avey et al. [21–23] presented the mathematical modeling for the stability problem of nanocomposite shell structures in a thermal environment and evaluated the influences of the distribution types and the volume fraction of CNTs on the buckling thermal and mechanical loads. The results show that the volume fraction of CNTs has an insignificant impact on the buckling loads, and the impact varies with the distribution types. In addition, the buckling behavior of FG-CNTRC truncated conical shells was examined by Eskandary et al. [24]. As for the buckling characteristics of functionally graded graphene platelet-reinforced composite (FG-GPLRC) truncated conical shells, Kiani [25] computed the critical thermal and mechanical loads and examined the influences of the mass volume of GPL, thermal environment, and shell length. It was revealed that the temperature elevation can decline the buckling pressure significantly. Using the trigonometric expansion, Mahani et al. [26] presented the thermal buckling analysis of FG-GPLRC conical shells. They found that the increasing mass volume of GPLs influences the exigent temperature. Li et al. [27] studied the thermal-elastic buckling of FG-GPLRC arch-shaped structures and analyzed the effects of geometric and material parameters of the structures on the critical temperature change. It was demonstrated that the critical temperature change increased significantly with the rising weight ratio of GPLs. Besides, Hasan et al. [28] and Ansari et al. [29], respectively, studied the thermal and mechanical post-buckling behaviors of FG-GPLRC conical shells.

The shells in the above-mentioned were treated as perfect structures without internal pores, but it is inevitable that pores may appear inside the composite materials [30]. Thus, the influence of internal pores on the static and dynamic behaviors of porous structures has received some researchers' attention [31–34]. Up to now, most of the research work focuses on porous cylindrical shells [35–37], and little has been conducted on porous conical shell structures. Hoa et al. [38] studied the buckling characteristics of FG porous truncated conical shells on Pasternak foundations and examined the effects of shell characteristics, internal pores, and foundations on the critical buckling load. Using the isogeometric analysis (IGA), Thanh et al. [39] obtained a three-dimensional solution of the buckling of FG porous-cellular conical shells to investigate the impact of pores on the critical buckling load. Nemati et al. [40] present a micromechanical model to study the stability of functionally graded conical panels resting on the Winkler–Pasternak foundation in various thermal environments. In their studies, the effects of the FGM profile, temperature distribution, and foundation parameters on the buckling thermal load are also investigated. In addition, Allahkaram et al. [41] and Fu et al. [42] analyzed the dynamic buckling behavior of porous truncated conical shells. From these studies on porous shell structures, the conclusion can be drawn that the internal pores weaken the stiffness of porous shells and decrease the buckling load.

As reviewed above, although porous FG-GPLRC shell structures can play a critical role in many applications, including aviation, rocket, shipbuilding, energy, and chemical engineering, the research work on the buckling behavior of porous FG-GPLRC truncated conical shells is relatively scarce. Moreover, the models for estimating the material properties of porous nanocomposites are complicated and based on the three assumptions about Young's elastic modulus, shear modulus, and mass density. Hence, this work attempts to present a refined model for estimating the material properties of porous nanocomposites and obtain the analytical solution of the buckling of the shells subjected to combined loading. The effects of the mass fraction and types of CNT distribution, the volume fraction and distribution of pores, and the geometric parameters of the shells on the critical buckling load are also analyzed.

2 A porous FG-GPLRC truncated conical shell

As depicted in Fig. 1, a porous FG-GPLRC truncated conical shell subjected to lateral pressure P_1, axial pressure P_2, and axial tension T is considered. The shell is assumed to be rested on the Winkler–Pasternak elastic foundation, and the foundation parameters are K_1 and K_2. The origin of a curvilinear coordinate system (x, θ, z) is placed on the cone top, and the x-θ plane is located on the middle surface of the shell. u, v, and w denote the corresponding displacements in the x-axis, θ-axis, and z-axis, respectively.s_1 and s_2 are the distances from the cone top to the small and large ends. L, h, and α imply the length, thickness, and semi-vertex. The

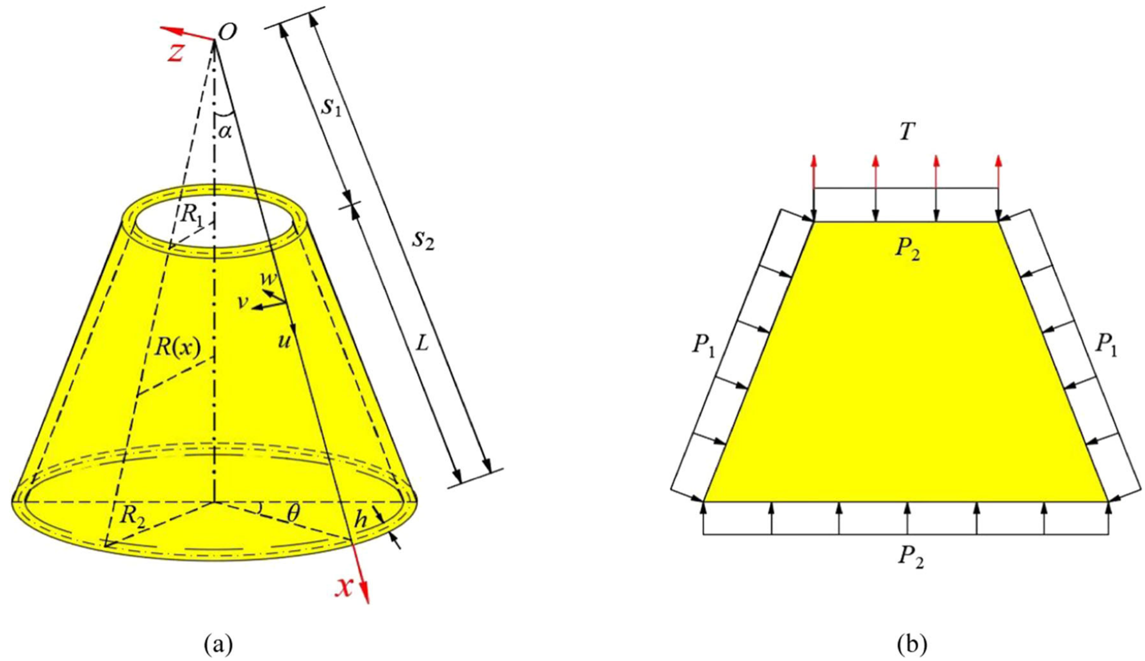


Fig. 1 A FG-GPLRC truncated conical shell: **a** geometrical parameters and **b** axial and lateral loads

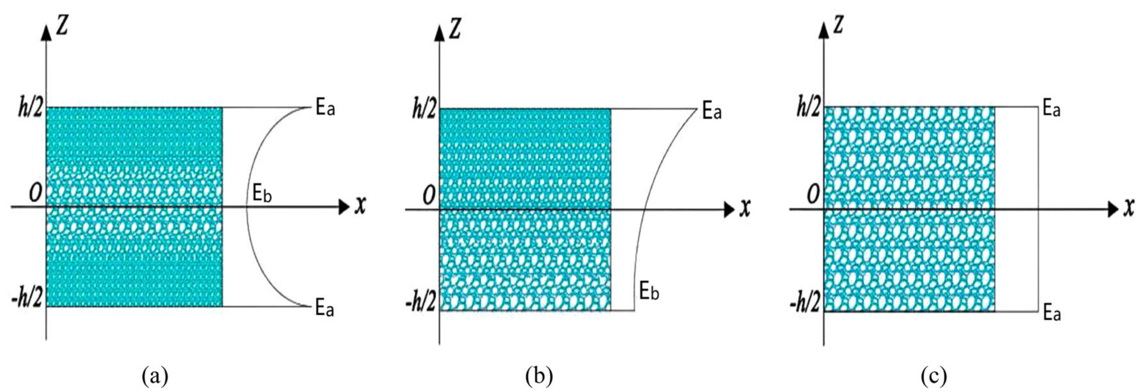


Fig. 2 Three types of porous distribution: **a** symmetric distribution (P-1), **b** asymmetric distribution (P-2), and **c** uneven distribution (P-3)

three types of porous distribution across the thickness, symbolized by "P-1", "P-2", and "P-3", are considered (Fig. 2). Apart from the other models of material properties [30, 31], which are based on Yang's modulus, mass density, and shear modulus, the present one is based on the porous volume fraction $V_{\mathrm{E}}(z)$ as

$$V_E(z) = e_0 \cos\left(\frac{\pi z}{h}\right) \quad (\mathrm{P}-1) \tag{1a}$$

$$V_E(z) = e_0 \cos\left(\frac{\pi z}{2h} + \frac{\pi}{4}\right) \quad (\mathrm{P}-2) \tag{1b}$$

$$V_E(z) = e_0 \quad (\mathrm{P}-3) \tag{1c}$$

where e_0 is the porous coefficient.

Young's elastic modulus $E(z)$ and the mass density $\rho(z)$ for various porous distributions are expressed in term of $V_{\mathrm{E}}(z)$[34];

$$E(z) = E_a(1 - V_E(z))$$
$$\rho(z) = \rho_a\left(1 - \frac{e_m}{e_0} V_E(z)\right) \tag{2}$$

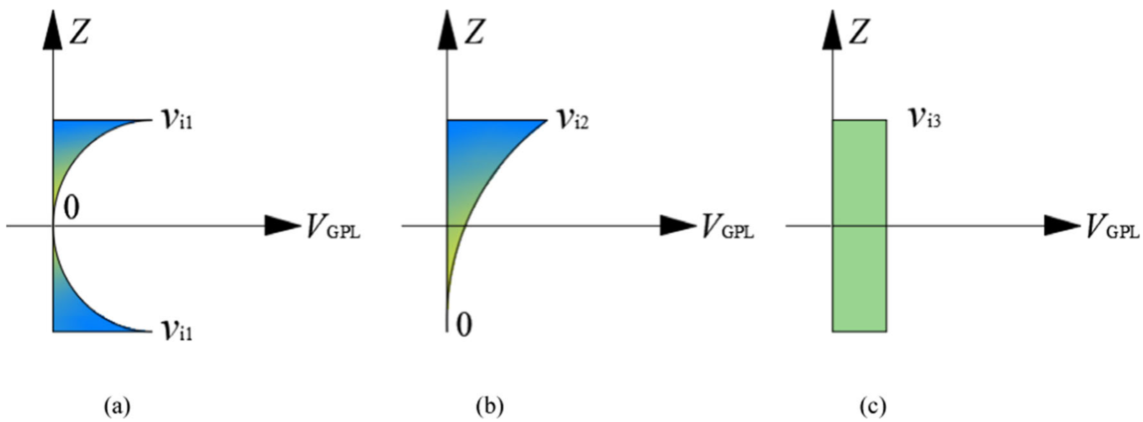


Fig. 3 Three patterns of GPL dispersion: **a** symmetric distribution (G-1), **b** asymmetric distribution (G-2), and **c** uneven distribution (G-3)

Here, e_m is the mass coefficient. E_a and ρ_a are the maximum values of Young's modulus and mass density, respectively.

The mass coefficient e_m and Poisson ratio $\mu(z)$ can be determined as follows [34]:

$$e_m = \frac{1.121 e_0 (1 - \sqrt[2.3]{1 - V_E(z)}}{V_E(z)}$$
$$\mu(z) = 0.121(\hat{p} + \mu_1 (0.342\hat{p}^2 - 1.21\hat{p} + 1.0)$$
$$\hat{p} = 1.121(1 - \sqrt[2.3]{1 - V_E(z)}) \tag{3}$$

Here, μ_1 denotes the Poisson ratio of the composites without pores.

As displayed in Fig. 3, three patterns of GPL dispersion, symbolized by "G-1", "G-2", and "G-3", are taken into account. The volume fraction $V_{\mathrm{GPL}}(z)$ of GPL for various distributions is assumed to be [34]

$$V_{\mathrm{GPL}}(z) = v_{i1}\left[1 - \cos\left(\frac{\pi z}{h}\right)\right] \quad (\mathrm{G} - 1) \tag{4a}$$

$$V_{\mathrm{GPL}}(z) = v_{i1}\left[1 - \cos\left(\frac{\pi z}{2h} + \frac{\pi}{4}\right)\right] \quad (\mathrm{G} - 2) \tag{4b}$$

$$V_{\mathrm{GPL}}(z) = v_{i3} \quad (\mathrm{G} - 3) \tag{4c}$$

Here, v_{i1},v_{i2} and v_{i3} are the maximum values for the three patterns of GPLs dispersion, determined by

$$\frac{W_{\mathrm{GPL}}}{W_{\mathrm{GPL}} + (\rho_{\mathrm{GPL}}/\rho_m)(1 - W_{\mathrm{GPL}})} \int_{-h/2}^{h/2} \left[1 - \frac{e_m}{e_0} V_{\mathrm{E}}(z)\right] \mathrm{d}z = \int_{-h/2}^{h/2} V_{\mathrm{GPL}}(z) \left[1 - \frac{e_m}{e_0} V_{\mathrm{E}}(z)\right] \mathrm{d}z \tag{5}$$

In Eqs. (1) and (2), the coefficients E_a,ρ_a, and μ_1 can be computed by using the Halpin–Tsai micromechanics model and the rule of mixture [34]:

$$E_a = \frac{3}{8}\left(\frac{1 + \xi_a \eta_a V_{\mathrm{GPL}}}{1 - \eta_a V_{\mathrm{GPL}}}\right) E_m + \frac{3}{8}\left(\frac{1 + \xi_b \eta_b V_{\mathrm{GPL}}}{1 - \eta_b V_{\mathrm{GPL}}}\right) E_m \tag{6}$$

$$\rho_a(z) = V_{\mathrm{GPL}}(z)\rho_{\mathrm{GPL}} + [1 - V_{\mathrm{GPL}}(z)]\rho_m \tag{7}$$

$$\mu_1(z) = V_{\mathrm{GPL}}(z)\mu_{\mathrm{GPL}} + [1 - V_{\mathrm{GPL}}(z)]\mu_m \tag{8}$$

in which

$$\xi_a = \frac{2a_{\mathrm{GPL}}}{h_{\mathrm{GPL}}}, \xi_b = \frac{2b_{\mathrm{GPL}}}{h_{\mathrm{GPL}}}, \eta_a = \frac{E_{\mathrm{GPL}}/E_m - 1}{E_{\mathrm{GPL}}/E_m + \xi_a}, \eta_b = \frac{E_{\mathrm{GPL}}/E_m - 1}{E_{\mathrm{GPL}}/E_M + \xi_b} \tag{9}$$

Here, E_{GPL},ρ_{GPL}, and μ_{GPL} imply Young's modulus, mass density, and the Poisson ratio of GPLs and E_{m},ρ_{m}, and μ_{m} are the corresponding values of the matrix. Besides, a_{GPL},b_{GPL}, and h_{GPL} denote the length, width, and thickness of GPLs.

3 Formulations

3.1 Governing equations

Due to the effects of lateral compressionP_1, axial compressionP_2, and axial tension T, the axial forces $\hat{N}_x$ and $\hat{N}_\theta$ and shear force $\hat{N}_{x\theta}$ can be calculated as [2, 8]

$$\hat{N}_x = 0.5\left(\frac{T}{x}S_2^2 - xP_1\right)\tan\alpha - \frac{P_2}{\pi x \sin 2\alpha}, \hat{N}_\theta = -xP_2\tan\alpha, \hat{N}_{x\theta} = 0 \tag{10}$$

Applying the classic thin shell theory and the Hamilton principle, after a long calculation, the static stability equations are expressed in terms of displacements u, v, and w as follows [8]:

$$\hat{L}_{11}(u) + \hat{L}_{12}(v) + \hat{L}_{13}(w) = 0 \tag{11a}$$

$$\hat{L}_{21}(u) + \hat{L}_{22}(v) + \hat{L}_{23}(w) = 0 \tag{11b}$$

$$\hat{L}_{31}(u) + \hat{L}_{32}(v) + \hat{L}_{33}(w) + P_1\hat{L}_{34}(w) + P_2\hat{L}_{35}(w) + T\hat{L}_{36}(w) = 0 \tag{11c}$$

Here, the operators $\hat{L}_{\mathrm{ij}}()$ are listed in Appendix.

3.2 The solution of the governing equations

In the present case, the small and large ends of the shell are assumed to be simply supported. The boundary conditions are stated as

$$v = w = M_1 = 0, \ \text{at} \ x = s_1, s_2 \tag{14}$$

The solution satisfying the boundary conditions is assumed to be [8]:

$$\begin{aligned} u &= u_{mn}\cos\frac{m\pi(x-s_1)}{L}\sin\left(\frac{n\theta}{2}\right) \\ v &= v_{mn}\sin\frac{m\pi(x-s_1)}{L}\cos\left(\frac{n\theta}{2}\right) \\ w &= w_{mn}\sin\frac{m\pi(x-s_1)}{L}\sin\left(\frac{n\theta}{2}\right) \end{aligned} \tag{15}$$

in which m and$n$ denote the half-wave numbers along the generatrix and the parallel circle.u_{mn},v_{mn}, and w_{mn} are the unknown amplitudes of displacements u, v, and w, respectively. To eliminating the space variables x and θ in Eqs. (11a)–(11b), the Galerkin method is used as follows [8]:

$$\begin{aligned} &\int_{s_1}^{s_2}\int_0^{2\pi}\Pi_1\cos\frac{m\pi(x-s_1)}{L}\sin\frac{n\theta}{2}x\sin\alpha\,\mathrm{d}x\mathrm{d}\theta = 0 \\ &\int_{s_1}^{s_2}\int_0^{2\pi}\Pi_2\sin\frac{m\pi(x-s_1)}{L}\cos\frac{n\theta}{2}x\sin\alpha\,\mathrm{d}x\,\mathrm{d}\theta = 0 \\ &\int_{s_1}^{s_2}\int_0^{2\pi}\Pi_3\sin\frac{m\pi(x-s_1)}{L}\sin\frac{n\theta}{2}x\sin\alpha\,\mathrm{d}x\,\mathrm{d}\theta = 0 \end{aligned} \tag{16}$$

in which

$$\begin{aligned} \Pi_1 &= x\left[L_{11}(u^{(1)}) + L_{12}(v^{(1)}) + L_{13}(w^{(1)})\right] \\ \Pi_2 &= x\left[L_{21}(u^{(1)}) + L_{22}(v^{(1)}) + L_{23}(w^{(1)})\right] \\ \Pi_3 &= x^2\left[L_{31}(u^{(1)}) + L_{32}(v^{(1)}) + L_{33}(w^{(1)})\right] \end{aligned} \tag{17}$$

Table 1 Comparison of the critical buckling hydrostatic pressure for isotropic truncated conical shells

α	$L/R1 = 0.5$			$L/R1 = 1.0$			$L/R1 = 2.0$		
	10^0	30^0	50^0	10^0	30^0	50^0	10^0	30^0	50^0
Baruch et al. [43]	19.40	14.55	8.813	8.569	5.843	3.285	3.740	2.237	1.164
Sofiyev [2]	19.373	14.397	8.640	8.534	5.666	3.098	3.688	2.077	1.016
Present	19.196	14.410	8.639	8.565	5.629	3.199	3.697	2.104	1.049

Table 2 Comparisons of the critical buckling hydrostatic pressure $\overline{P}_{\mathrm{Hcr}}$ and axial tension $\overline{T}\mathrm{cr}$ for FGM truncated conical shells

Index N	$\overline{P}_{\mathrm{Hcr}}$ (MPa)			$\overline{T}\mathrm{cr}$ (MPa)		
	$N = 0.0$	$N = 1.0$	$N = 2.0$	$N = 0.0$	$N = 1.0$	$N = 2.0$
Sofiyev [2]	0.2993	0.1667	0.1573	14.228	7.923	7.448
Present	0.3011	0.1689	0.1586	14.401	7.952	7.463

Substituting Eqs. (16) and (17) into (11a)–(11c), an algebraic equation can be derived as

$$(a_1 + a_2 P_1 + a_3 P_2 + a_4 T) w_{mn} = 0 \tag{18}$$

Here, $a_i (i = 1–4)$ are integral coefficients.

According to Eq. (18), the buckling axial pressure ($P_1 = T = 0$, $P_2 \neq 0$), hydrostatic pressure ($T = 0$, $P_1 = P_2 = P_{\mathrm{H}} \neq 0$), and axial tension ($P_{\mathrm{H}} = 0$, $T \neq 0$) can be obtained $\overline{P}_{\mathrm{a}} = -a_1/a_3$, $\overline{P}_{\mathrm{H}} = -a_1/(a_2 + a_3)$, and $\overline{T}_{\mathrm{cr}} = -a_1/a_4$.

If the half-vertex $\alpha \to 0$, the truncated conical shell can be transformed into the corresponding cylindrical shell. The following expressions are applied in computing the buckling load of the cylindrical shell [2]:

$\alpha \to 0, S_1 \to \infty, S_1 \sin\alpha = R, \ln(S_1/S_2)\sin\alpha = R/L.$

4 Results and discussion

4.1 Comparison studies

In this subsection, the following numerical examples are presented to validate the accuracy of the present method:

Example 1 In Table 1, the dimensionless buckling hydrostatic pressure $P_{\mathrm{Hcr}} = \overline{P}_{\mathrm{H}}/E \times 10^6$ of an isotropic truncated conical shell is listed and compared with those given by Sofiyev [2] and Baurch et al. [43]. The radius–thickness ratio R_1/h and Poisson ratio μ are 100 and 0.3, respectively. The results show that the present results agree well with those in the open literature.

Example 2 The buckling hydrostatic pressure $\overline{P}_{\mathrm{Hcr}}$ and axial tension $\overline{T}\mathrm{cr}$ of an FGM truncated conical shell made of ceramic and metal are calculated and displayed in Table 2. Young's elastic modulus and Poisson ratio are $E(z) = (E_c - E_m)\left(\frac{1}{2} + \frac{z}{2h}\right)^N + E_m$ and $\mu(z) = (\mu_c - \mu_m)\left(\frac{1}{2} + \frac{z}{2h}\right)^N + \mu_m$, in which E_c and μ_c are Young's elastic modulus and Poisson ratio of the ceramic, and E_m and μ_{m} denote the corresponding values of the metal. The material and geometrical parameters are given by Sofiyev [2]. It can be observed that the present results are in good agreement with those obtained by Sofiyev [2].

Example 3 Table 3 shows the comparison of the buckling load ratio $N_{\mathrm{GPL}}/N_{\mathrm{polymer}}$ for a FG-GPLRC cylindrical shell. Young's modulus and Poisson ratio are $E_{\mathrm{m}} = 3.0\mathrm{GPa}, \nu_m = 0.34, E_{\mathrm{GPL}} = 1.01\mathrm{TPa}$, and $\nu_{\mathrm{GPL}} = 0.186$. The geometrical parameters are $a_{\mathrm{GPL}} = 2.5\mu\mathrm{m}, b_{GPL} = 1.5\mu\mathrm{m}, h_{\mathrm{GPL}} = 1.5\mathrm{nm}$, $R = 200\mathrm{mm}$, $h = 4\mathrm{mm}$, and $L = 1000\mathrm{mm}$. The mass fraction of GPLs is $W_{\mathrm{GPL}} = 0.5\%$ and the four types of GPL dispersion UD, FGA, FGO, and FGX are described by Ansari et al. [29] and Wang et al. [44]. It is clear that the present results are close to those presented by Ansari et al. [29] and Wang et al. [44].

Table 3 Comparison of the buckling load ratio for FG-GPLRC truncated cylindrical shells subjected to hydrostatic pressure

Type	UD	FGA	FGO	FGX
Wang et al. [44]	2.7409	2.5099	2.1792	3.1961
Ansari et al. [29]	2.7371	2.5543	2.2677	3.1283
Present	2.7384	2.5512	2.2417	3.1326

Table 4 Comparison of the critical buckling load for an isotropic porous truncated conical shells rested on Winkler–Pasternak foundation

e_0	0.2	0.3	0.4	0.5	0.6
Hoa et al. [38]	388.294	358.551	327.929	296.196	262.998
Present	384.411	354.208	324.650	292.319	259.053
Error	1.0%	1.2%	1.0%	1.3%	1.5%

Example 4 Table 4 shows the comparison of the critical axial compression P^* for an isotropic porous truncated conical shell rested on the Winkler–Pasternak foundation. The pores are uniformly distributed in the shell (P-3). The parameters of material, geometry, and elastic foundation are $E = 200.1\text{GPa}, \nu = 0.34$, $h = 25\text{mm}$, $R_1 = 40h$, $L = 2R_1, \alpha = 15^{\circ}$, and $K_1 = 2.5 \times 10^7 \text{N/m}^2$, $K_2 = 2.5 \times 10^5 \text{N/m}^2$. The critical compression P^* is defined as $P^* = \overline{P}_{\text{acr}}/P_{\text{cl}}$, where $P_{\text{acr}} = 2\pi E h^2 \cos^2\alpha/\sqrt{3(1-\nu^2)}$. It is seen that the present results are close to those presented by Hoa et al. [38]. The maximum error is 1.5%. That is caused by the different shell theory. The theory used by Hoa et al. [38] is the shear deformation theory. However, the present theory is the classic thin shell theory.

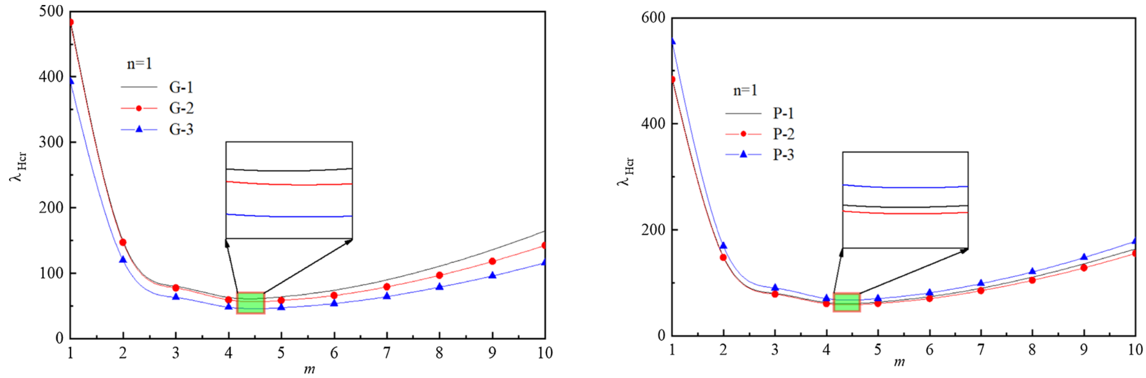
4.2 Parametric studies

After the present method is validated, the effects of GPLs, pores, and geometrical parameters on the buckling pressure and axial tension for a porous FG-GPLRC truncated conical shell are examined in this subsection. The material and geometric parameters of GPLs are the same as those given in Example 3: Young's modulus and Poisson ratio of the matrix are $E_m = 130$ GPa and $\nu_{\text{m}} = 0.34$. The dimensionless buckling hydrostatic pressure and axial tension are $\lambda_{\text{Hcr}} = \overline{P}_H/E_m \times 10^6, \lambda_{\text{Tcr}} = \overline{T}/E_m \times 10^6$. Unless there is a specific statement, the distribution patterns of GPLs and porous are G-1 and P-1. The other calculating data are $h = 0.01$ mm, $R_1/h = 100$, $L/R_1 = 1.0$, and $\alpha = 30^o$.

For the sake of detecting the critical buckling mode (m,n), Figs. 4, 5, 6, and 7 show the variation of the buckling hydrostatic pressure λ_{Hcr} and axial tension λ_{Tcr} with the generatrix half-wave number m and circumferential half-wave number n. The three patterns of GPL dispersion and three types of porous distribution are considered. The mass volume W_{GPL} of GPL and porous coefficient e_0 are 0.5% and 0.2, respectively. It is found that the buckling pressure and tension are not monotonously varied with the half-wave numbers m and n. When the mode (m,n) is (1,16), both the buckling hydrostatic pressure and axial tension get the minimum values. Hence, (1,16) is the critical buckling mode and the following buckling hydrostatic pressure and axial tension are obtained at the mode. From Figs. 4, 5, 6, and 7, it was also found the buckling load for Type P-1 are larger than those for Types P-2 and P-3. It is because most pores for Type P-1 appear in the middle of the shell. Hence, the effective stiffness for Type P-1 is larger than those for Types P-2 and P-3.

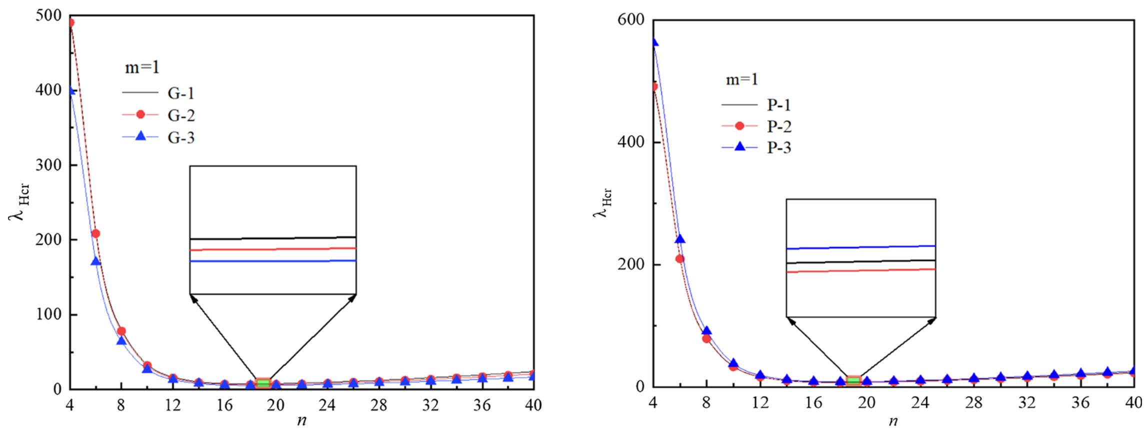
Figures 8 and 9 reveal that the mass fraction W_{GPL} and the three patterns of GPL dispersion have significant effects on the critical buckling pressure λ_{Hcr} and axial tension λ_{Tcr}. Both the critical pressure and axial tension increase, and the mass volume W_{GPL} rises. This is due to the fact that Young's modulus of GPL is significantly higher than that of the matrix. Because more GPLs for Pattern G-1 are implanted in the inner and outer surfaces of the shell, the effective stiffness for Pattern G-1 is evidently larger than those for Patterns G-2 and G-3. Hence, the increasing speed for Pattern G-1 is higher than those for Patterns G-2 and G-3. When the value of W_{GPL} changes from 0 to 0.05,λ_{Hcr} and λ_{Tcr} for Pattern G-1 increase by about 323.3% and 297.2%, respectively. However,λ_{Hcr} and λ_{Tcr} for Pattern G-3 only increase by 33.8% and 33.7%. Thus, Pattern G-3 may be used to obtain the higher stiffness of the porous shell.

The influences of the porous coefficient e_0 and the three types of porous distributions on the critical buckling pressure λ_{Hcr} and axial tension λ_{Tcr} are shown in Figs. 10 and 11. As expected, because internal pores weaken



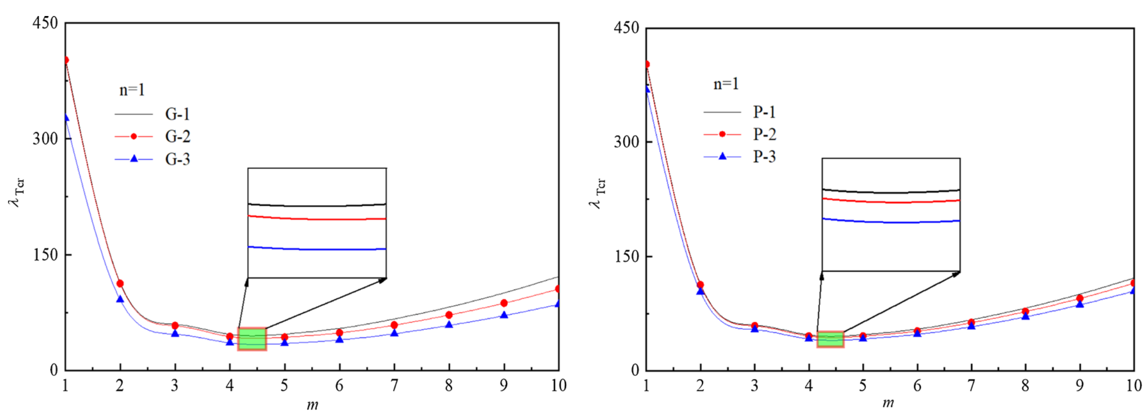
(a) (b)

Fig. 4 Variation of buckling hydrostatic pressure with the half-wave number *m* for **a** various GPL patterns and **b** various porous distributions



(a) (b)

Fig. 5 Variation of buckling hydrostatic pressure with the half-wave number *n* for **a** various GPL patterns and **b** various porous distributions



(a) (b)

Fig. 6 Variation of buckling axial tension with the half-wave number m for **a** various GPL patterns and **b** various porous distributions

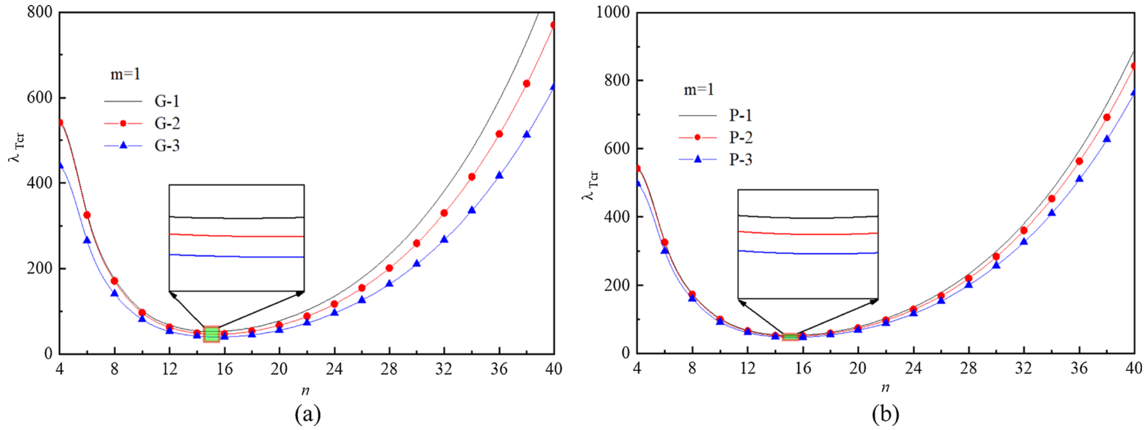


Fig. 7 Variation of buckling axial tension with the half-wave number n for **a** various GPL patterns and **b** various porous distributions

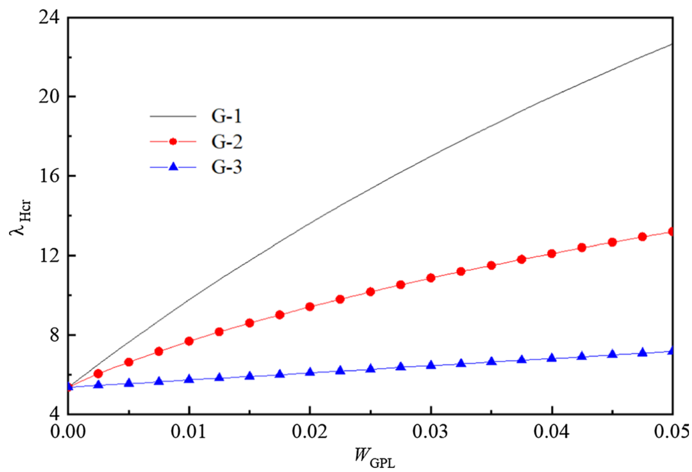


Fig. 8 Effect of GPL on the buckling hydrostatic pressure

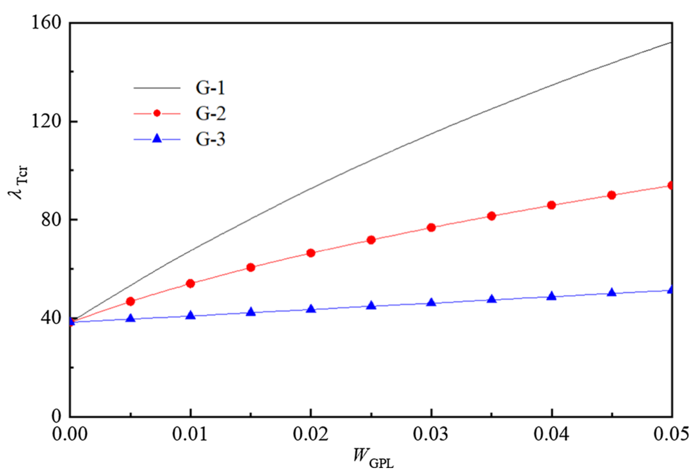


Fig. 9 Effect of GPL on the buckling axial tension

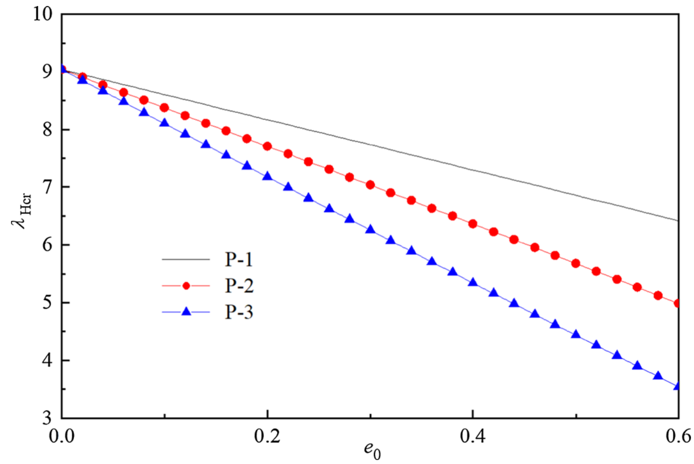


Fig. 10 Effect of pores on the buckling hydrostatic pressure

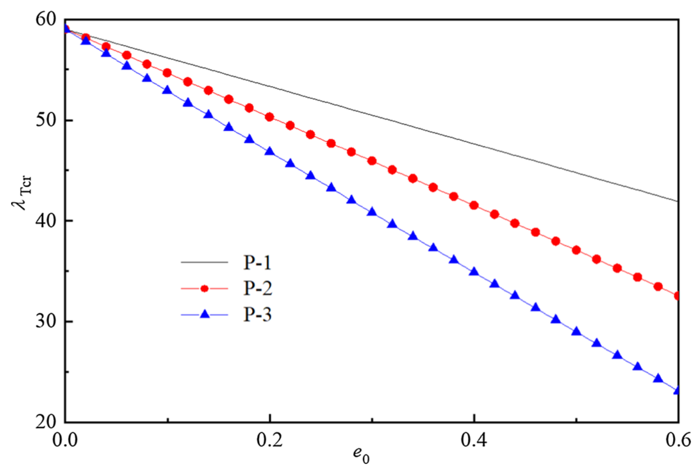


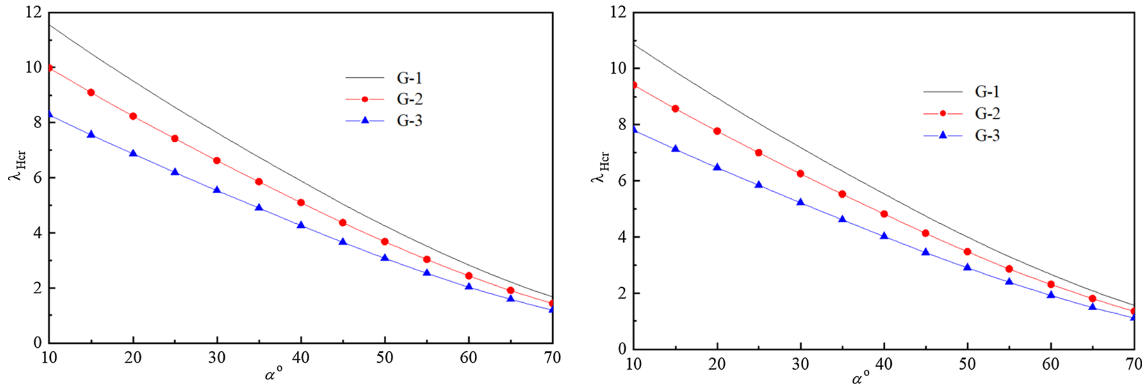
Fig. 11 Effect of pores on the buckling axial tension

the effective stiffness of the shell, both the critical pressure and axial tension decline with the increase in the porous coefficient. The critical pressure and tension for Type P-3 decline faster than those for Types P-1 and P-2. When the value of e_0 increases from 0.0 to 0.6, λ_{Hcr} and λ_{Tcr} for Type P-2 decrease by about 60.9% and 60.8%, and those for Type P-1 reduce about 29.0% and 29.1. Therefore, the pores for Type P-1 should be avoided in the fabricating of the porous shell.

Figures 12 and 13 show the effect of the semi-vertex α on the critical buckling pressure λ_{Hcr} and axial tension λ_{Tcr} of the shell distributed with the pores for Types P-1 and P-2. It is seen that λ_{Hcr} and λ_{Tcr} can be raised by increasing the semi-vertex. Among the three patterns of GPL dispersion,λ_{Hcr} and λ_{Tcr} for Type G-1 decrease faster than those for G-2 and G-3.

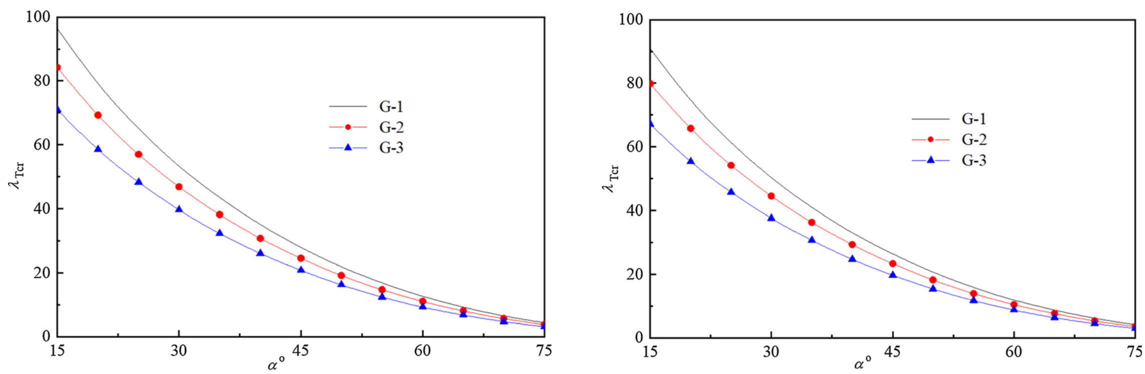
The effect of the length–radius ratioL/R_1 on the critical buckling pressure λ_{Hcr} for different types of GPL and pores is revealed in Fig. 14. It is seen that the critical buckling pressure is decreased with the increase in the ratio. When the value of the ratioL/R rises from 0.5 to 2.0, λ_{Hcr} reduces by about 86.3%. That is because the stability of a stubby conical shell is better than that of a slender one.

Figure 15 illustrates the effect of the radius–thickness ratio R_1/h on the critical buckling pressure λ_{Hcr}. As expected,λ_{Hcr} for a thick conical shell is higher than that for the corresponding thin shell. As the value of R_1/h increases from 100 to 300,λ_{Hcr} decreases by about 88.5%. This is due to the fact that thick shells have higher stiffness than the corresponding thin shells.



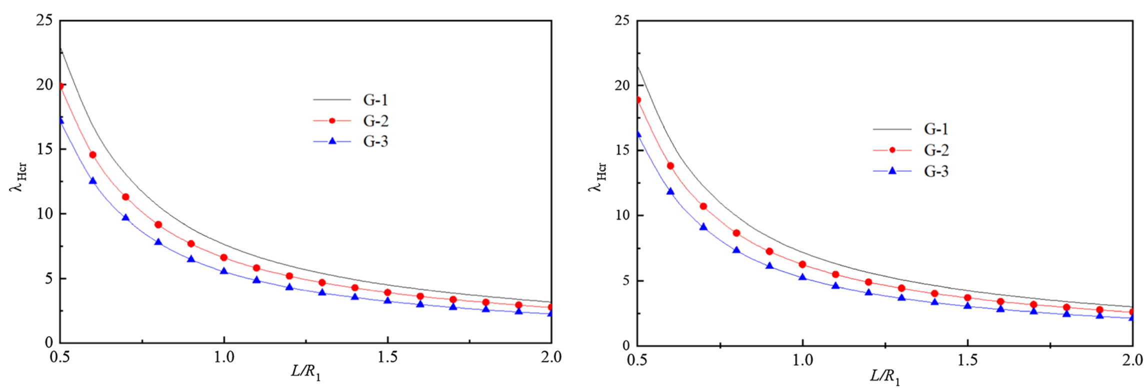
(a) (b)

Fig. 12 Effect of half-vertex on the buckling hydrostatic pressure: **a** Type P-1 and **b** Type P-2



(a) (b)

Fig. 13 Effect of half-vertex on the buckling axial tension: **a** Type P-1 and **b** Type P-2

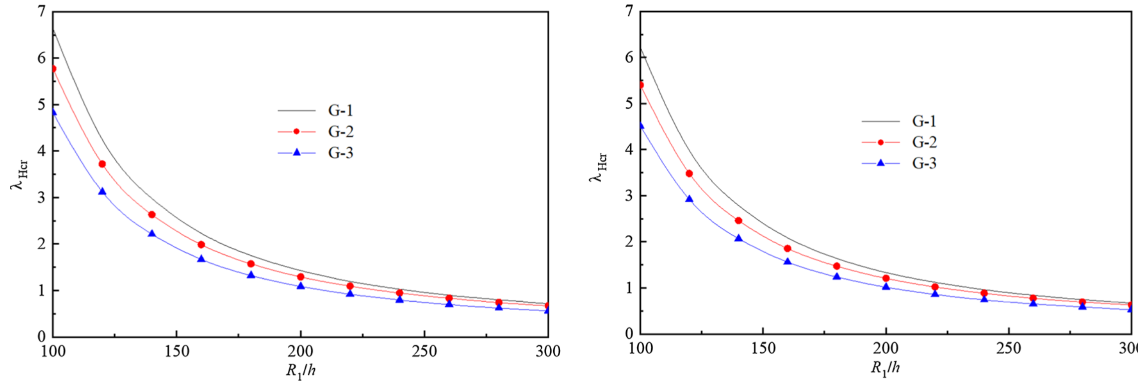


(a) (b)

Fig. 14 Effect of length–radius ratio on the buckling hydrostatic pressure: **a** Type P-1 and **b** Type P-2

5 Conclusion

In this study, a reliable and effective method to investigate the buckling behavior of porous FG-GPRC truncated conical shells subjected to hydrostatic pressure and axial tension is presented. A new refined model for evaluating the material properties of porous nanocomposites is presented, and the analytical solution of the buckling hydrostatic pressure and axial tension is obtained. The effects of pores, GPLs, and shell geometrical



(a) (b)

Fig. 15 Effect of radius–thickness ratio on the buckling hydrostatic pressure: **a** Type P-1 and **b** Type P-2

parameters on the buckling pressure and axial tension are examined in detail. Some novel conclusions are drawn as follows:

1. As the porous coefficient, semi-vertex, and length–thickness ratio rise, the critical buckling pressure and axial tension decline. On the contrary, the critical pressure and tension are raised with the increase in the mass volume of GPLs.
2. The patterns of GPL dispersion have a significant influence on the critical buckling pressure and the axial tension. Furthermore, the increasing speed for Pattern G-1 is higher than those for G-2 and G-3. If the value W_{GPL} of changes from 0 to 0.05,λ_{Hcr} and λ_{Tcr} for Pattern G-2 increase by about 323.3% and 297.2%. However,λ_{Hcr} and λ_{Tcr} for Pattern G-3 only rise by 33.8% and 33.7%.
3. The types of porous distributions have an evident effect on the critical buckling pressure and the axial tension. The critical pressure and tension for Type P-3 reduce faster than those for Types P-1 and P-2.
4. The buckling hydrostatic pressure λ_{Hcr} declines as the length–radius ratio L/R_1 increases. If the value of the ratio rises from 0.5 to 2.0,λ_{Hcr} reduces by about 86.3%.
5. The critical buckling pressure for a thin truncated conical shell is smaller than that for the corresponding thick shell. When the value of the length–radius ratio increases from 100 to 300, the buckling pressure declines by about 88.5%.

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Author contributions X.L. conceived this study and conducted manuscript writing. W.J. conducted table and graphic output of data. W.Y. verified the content of the article. W.W. participated in the drafting of the manuscript and made improvements.

Declarations

Competing interests The authors declare no competing interests.

Appendix

$$L_{11}() = A_{11}x\frac{\partial^2()}{\partial x^2} + A_{11}\frac{\partial()}{\partial x} - A_{22}\frac{()}{x} + \frac{1}{x\sin^2\alpha}A_{66}\frac{\partial^2()}{\partial\theta^2}$$

$$L_{12}() = (A_{12} + A_{66})\frac{1}{\sin\alpha}\frac{\partial^2()}{\partial x\partial\theta} - (A_{22} + A_{66})\frac{1}{x\sin\alpha}\frac{\partial()}{\partial\theta}$$

$$L_{13}() = -B_{11}x\frac{\partial^3()}{\partial x^3} - (B_{12} + 2B_{66})\frac{1}{x\sin^2\alpha}\frac{\partial^3()}{\partial x\partial\theta^3} - B_{11}\frac{\partial^2()}{\partial x^2}$$

$$+ (B_{12} + B_{22} + 2B_{66})\frac{1}{x^2\sin^2\alpha}\frac{\partial^2()}{\partial\theta^2} + A_{12}\cot\alpha\frac{\partial()}{\partial x}$$
$$+ B_{22}\frac{1}{x}\frac{\partial()}{\partial x} - A_{22}\cot\alpha\frac{1}{x}()$$

$$L_{21}() = (A_{12} + A_{66})\frac{1}{\sin\alpha}\frac{\partial^2()}{\partial x\partial\theta} + (A_{22} + A_{66})\frac{1}{x\sin\alpha}\frac{\partial()}{\partial\theta}$$

$$L_{22}() = A_{22}\frac{1}{x\sin^2\alpha}\frac{\partial^2()}{\partial\theta^2} + A_{66}x\frac{\partial^2()}{\partial x^2} + A_{66}\frac{\partial()}{\partial x} - \frac{1}{x}A_{66}()$$

$$L_{23}() = -(B_{12} + 2B_{66})\frac{1}{\sin\alpha}\frac{\partial^3()}{\partial x^2\partial\theta} - B_{22}\frac{1}{x^2\sin^2\alpha}\frac{\partial^3()}{\partial\theta^3}$$
$$- B_{22}\frac{1}{x\sin\alpha}\frac{\partial^2()}{\partial x\partial\theta} + A_{22}\cot\alpha\frac{1}{x\sin\alpha}\frac{\partial()}{\partial\theta}$$

$$L_{31}() = B_{11}x\frac{\partial^3()}{\partial x^3} + (B_{12} + 2B_{66})\frac{1}{x\sin^2\alpha}\frac{\partial^3()}{\partial x\partial\theta^2} + 2B_{11}\frac{\partial^2()}{\partial x^2}$$
$$+ B_{22}\frac{1}{x^2\sin^2\alpha}\frac{\partial^2()}{\partial\theta^2} - (A_{12}\cot\alpha + \frac{1}{x}B_{22})\frac{\partial()}{\partial x} + \frac{1}{x^2}B_{22}() - A_{22}\cot\alpha\frac{1}{x}()$$

$$L_{32}() = (B_{11} + 2B_{66})\frac{1}{\sin\alpha}\frac{\partial^3()}{\partial x^2\partial\theta} + B_{22}\frac{1}{x^2\sin^3\alpha}\frac{\partial^3()}{\partial\theta^3}$$
$$- B_{22}\frac{1}{x\sin\alpha}\frac{\partial^2()}{\partial x\partial\theta} + \left(B_{22}\frac{1}{x^2\sin\alpha} - A_{22}\cot\alpha\frac{1}{x\sin\alpha}\right)\frac{\partial()}{\partial\theta}$$

$$L_{33}() = -D_{11}x\frac{\partial^4()}{\partial x^4} - D_{22}\frac{1}{x^3\sin^4\alpha}\frac{\partial^4()}{\partial\theta^4} - (D_{12} + 2D_{66})\frac{2}{x\sin^2\alpha}\frac{\partial^4()}{\partial x^2\partial\theta^2}$$
$$+ (D_{12} + 2D_{66})\frac{2}{x^2\sin^2\alpha}\frac{\partial^3()}{\partial x\partial\theta^2} - 2D_{11}\frac{\partial^3()}{\partial x^3} + \left(\frac{1}{x}D_{22} + 2B_{12}\cot\alpha + xK_2\right)\frac{\partial^2()}{\partial x^2}$$
$$+ \left[B_{22}\text{cot}\alpha\frac{2}{x^2\sin^2\alpha} - (D_{12} + 2D_{66} + D_{22})\frac{2}{x^3\sin^2\alpha} + \frac{K_2}{x\sin\alpha}\right]\frac{\partial^2()}{\partial\theta^2}$$
$$K_2\frac{\partial()}{\partial x} - D_{22}\frac{1}{x^2}\frac{\partial()}{\partial x} + B_{22}\cot\alpha\frac{1}{x^2}() - A_{22}\cot^2\alpha\frac{1}{x}() - xK_1()$$

$$L_{34}() = -\frac{1}{2}\left(x\frac{\partial()}{\partial x} + x^2\frac{\partial^2()}{\partial x^2}\right)$$

$$L_{35}() = -\frac{2}{\sin 2\alpha}\frac{\partial^2()}{\partial\theta^2}$$

$$L_{36}() = \frac{1}{2}s_1^2\tan\alpha\frac{\partial^2()}{\partial x^2}$$

where the stiffness coefficients A_{ij}, B_{ij} and D_{ij} can be calculated by

$$(A_{11}, B_{11}, D_{11})^T = (A_{22}, B_{22}, D_{22})^T = \int_{-0.5h}^{0.5h}(1, z, z^2)\frac{E(z)}{1-\mu^2(z)}\mathrm{d}z$$

$$(A_{12}, B_{12}, D_{12})^T = (A_{21}, B_{21}, D_{21})^T = \int_{-0.5h}^{0.5h}(1, z, z^2)\frac{\mu(z)E(z)}{1-\mu^2(z)}\mathrm{d}z$$

$$(A_{66}, B_{66}, D_{66})^T = \int_{-0.5h}^{0.5h}(1, z, z^2)\frac{E(z)}{2[1+\mu(z)]}\,\mathrm{d}z$$

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